Agentic Delegation and the Language Frontier

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https://github.com/wbgradost/ai-03-quispe

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Citation audit

Baseline: inferred Quispe, Claude Code and expansion beyond prior training. Verified v2: Alexander Quispe and Kevin Xu; *Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub*; v1 May 25, v2 July 7, 2026. The evidence is an observational developer-month panel, not an experiment.

Agent problem and activation band

$$V^g = \max_{m \in \mathcal{M}_g} V^m, \quad Z^g = \mathbf{1}\{V^g \geq 0\}, \quad \mathcal{M}_1 = \{S, C\} \subset \mathcal{M}_2 = \{S, C, D\}.$$

For unfamiliar languages, conversational help leaves T^S unchanged. If $B = T^S - T^D > 0$,

$$Z^2 - Z^1 = \mathbf{1}\{T^D \leq \omega < T^S\}.$$

Delegation makes the middle band feasible. This expands a production frontier, not necessarily human skill.

Results and conditions

- Menu inclusion: $Z^2 \geq Z^1$, $N^2 \geq N^1$ path by path.
- Activation band: unfamiliar language, Assumption 1, $B > 0$.
- Continuous F: activation probability $F(T^S) - F(T^D)$.
- Cumulative effect nonnegative for $0 \leq p^1 \leq p^2 \leq 1$.
- Strict expansion requires positive mass and a nonempty band.

Independent check: Proposition 3

For $p^1 = 0$, $\Delta(s) = 1 - (1 - p^2)^{s+1}$ and

$$\Delta(s+1) - \Delta(s) = p^2(1 - p^2)^{s+1}.$$

At the permitted endpoint $p^2 = 1$, $\Delta(s) = 1$ for all $s \geq 0$. The printed strict-growth/strict-concavity claim fails there. Correct strict domain: $0 < p^2 < 1$, with a nonempty relevant candidate set.

Evidence and causal limit

5,346 developers, 149,688 developer-months, 57.2 million changed files. Adoption-month estimates: languages +2.53, new languages +1.19, entropy +0.382, cumulative breadth +1.604. Robust event-time association, but voluntary adoption may coincide with an unfamiliar-language project shock; causality is not settled.

Frontier expansion versus deskilling

Quispe–Xu: AI adds delegated execution and lowers language-specific entry thresholds (extensive margin). Aouad–Lykouris–Zhong: AI, skill and effort are perfectly substitutable, so AI can crowd out effort (intensive margin).

Hand derivation: Proposition 3 endpoint

P3: comprobación del caso extremo $p^2=1$

$$C(s) = (1-p^1)^{s+1} - (1-p^2)^{s+1}$$

$$\downarrow \; p^1 = 0 < p^2$$

$$C(s) = 1-(1-p^2)^{s+1}$$

$$C(s+1) - C(s) = [1-(1-p^2)^{s+2}] - [1-(1-p^2)^{s+1}]$$

$$= (1-p^2)^{s+1} - (1-p^2)^{s+2}$$

$$= p^2(1-p^2)^{s+1}$$

Para $p^2(1-p^2)^{s+1} > 0$: $0 < p^2 < 1 \neq p^2 = 1$

$$C(s+1) - C(s) = 1(1-1)^{s+1} = 0$$

Falla cuando $p^2 = 1$

Lean pipeline and fidelity verdict

PAPER STATEMENT → LEAN STATEMENT → PROOF → FIDELITY CHECK → VERDICT

- WSL verified Lean 4.30.0-rc2 and Lake; exact arXiv v2 source was privately hash-pinned.
- The official scaffold failed before creating the paper folder because the root EconCSLib module was not yet built.
- The base build was stopped at closeout; all proof targets remain open and the fast check fails because `status.json` is absent.
- Lean compilation and faithful representation are distinct: neither is claimed complete here.
- Independent verdict: the printed strict clause fails at $p^2 = 1$; an interior condition $0 < p^2 < 1$ is necessary.

Status: **PARTIAL**.